Technical Supplement

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A Beyond Established FND Benchmarks: Challenges in Early Few-Shot Detection

Table 1 provides detailed in-domain and cross-domain results for all models.

B Implementation Details

B.1 Prompt Templates

Verifier prompt with article and selected sentences. After the Selector produces S^*, the Verifier receives the full article, publication date, and selected sentences subset. The template is adapted from the default template of Jin et al. [4] and adjusted for the FND task.

System message.

> You are an expert fact-checker who verifies news articles using a search tool.
> Here are instructions: 1. Read the entire article and the identified suspicious sentences. 2. Think step by step before calling the tool and after you receive the result of the tool call. Decide what query to call the tool with. 3. Call the tool with the query you have decided on. If the returned results are empty, revise and retry the query. 4. Think step by step again after you receive the result of the tool call. If you have the information you need, you can stop here. 5. Otherwise, come up with a new query that combines information from the previous results. 6. Include your final answer inside `<answer>` ... `</answer>`. The answer should be either *real* or *fake*; for example, `<answer> real </answer>` or `<answer> fake </answer>`.

User message.

> Decide whether a news article is real or fake based on the provided article and the identified suspicious sentences. The title should be treated as part of the article. Now is `{PUBLISHED_DATE}`, which is the same date as the publication date of the news. You must conduct reasoning inside `<think>` and `</think>` first every time you get new information. After reasoning, if you find you lack some knowledge or need to access real-time data, you can call a search engine by `<search> query </search>`, and it will return the top searched results between `<information>` and `</information>`. You can search as many times as you want. If you find no further external knowledge needed, you can directly provide the answer without detailed illustrations. For example, `<answer> real </answer>` or `<answer> fake </answer>`. Question: Is the news real or fake? Article: `{ARTICLE}` (Published date: `{PUBLISHED_DATE}`). Suspicious Sentences: `{SELECTED_SENTENCES}`.

Verifier prompt with article only. For model training without the Selector, the prompt uses the same search and answer format but omits selected evidence.

System message.

> You are an expert fact-checker who verifies news articles using a search tool.
> Here are instructions: 1. Read the entire article. 2. Think step by step before calling the tool and after you receive the result of the tool call. Decide what query to call the tool with. 3. Call the tool with the query you have decided on. If the returned results are empty, revise and retry the query. 4. Think step by step again after you receive the result of the tool call. If you have the information you need, you can stop here. 5. Otherwise, come up with a new query that combines information from the previous results. 6. Include your final answer inside `<answer>` ... `</answer>`. The answer should be either *real* or *fake*; for example, `<answer> real </answer>` or `<answer> fake </answer>`.

User message.

> Decide whether a news article is real or fake based on the provided article and the identified suspicious sentences. The title should be treated as part of the article. Now is `{PUBLISHED_DATE}`, which is the same date as the publication date of the news. You must conduct reasoning inside `<think>` and `</think>` first every time you get new information. After reasoning, if you find you lack some knowledge or need to access real-time data, you can call a search engine by `<search> query </search>`, and it will return the top searched results between `<information>` and `</information>`. You can search as many times as you want. If you find no further external knowledge needed, you can directly provide the answer without detailed illustrations. For example, `<answer> real </answer>` or `<answer> fake </answer>`. Question: Is the news real or fake? Article: `{ARTICLE}` (Published date: `{PUBLISHED_DATE}`).

Query-focused summarizer prompt. Retrieved passages are summarized before being returned to the Verifier. The summarizer receives the news article, current query, and retrieved documents, and is prompted as follows:

> **Task:** Synthesize the key information from the retrieved documents that is relevant to the current query, in order to support downstream fact-checking of the news article.
> **Instructions:** Extract and merge relevant facts such as dates, events, statistics, direct statements, and attributed claims; eliminate redundancy; and provide information for fact-checking the news article rather than merely answering the current query. If the documents do not contain relevant information, state that the sources are insufficient and suggest further investigation directions or query rewrites. Return only the synthesized information, with no additional text.
> **Context:** News Article: `{ARTICLE}`. Current Query: `{QUERY}`. Retrieved Documents: `{DOCUMENTS}`.

B.2 Search Corpus Preparation

For each article, we construct an instance-specific retrieval corpus to support the Verifier [1]. Given the article title, full content, and

[1] We use a local retriever for cost control. In GRPO training, each prompt samples G trajectories and each trajectory can invoke up to B search actions, leading to up

Table 1: In-domain and transfer results on PolitiFact, GossipCop, and BreakNews.

Dataset	Cat.	Model	Source				Zero-shot (BreakNews)				Few-shot (BreakNews)			
			Acc	Prec	Rec	F1	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
PolitiFact	(i)	DeClarE [7]	0.658	0.667	0.663	0.657	0.512	0.512	0.512	0.512	$0.503_{(0.001)}$	$0.566_{(0.017)}$	$0.503_{(0.001)}$	$0.349_{(0.004)}$
		MAC [10]	0.725	0.726	0.722	0.723	0.490	0.489	0.490	0.483	$0.522_{(0.028)}$	$0.520_{(0.029)}$	$0.520_{(0.028)}$	$0.518_{(0.028)}$
		GET [12]	0.719	0.720	0.720	0.719	0.482	0.482	0.482	0.480	$0.513_{(0.044)}$	$0.511_{(0.046)}$	$0.511_{(0.045)}$	$0.508_{(0.044)}$
		MUSER [5]	0.700	0.708	0.704	0.699	0.496	0.496	0.496	0.494	$0.571_{(0.027)}$	$0.573_{(0.027)}$	$0.571_{(0.027)}$	$0.567_{(0.029)}$
		CompareNet [3]	0.704	0.704	0.698	0.699	0.491	0.366	0.491	0.407	$0.493_{(0.019)}$	$0.494_{(0.020)}$	$0.493_{(0.019)}$	$0.489_{(0.016)}$
	(ii)	dEFEND [9]	0.733	0.733	0.732	0.732	0.458	0.457	0.458	0.455	$0.491_{(0.013)}$	$0.308_{(0.082)}$	$0.491_{(0.013)}$	$0.347_{(0.019)}$
		WSDMS [13]	0.783	0.787	0.786	0.783	0.482	0.481	0.482	0.473	$0.534_{(0.042)}$	$0.535_{(0.043)}$	$0.534_{(0.042)}$	$0.531_{(0.042)}$
	(iii)	GenFEND [6]	0.842	0.841	0.842	0.841	0.482	0.481	0.482	0.476	$0.478_{(0.034)}$	$0.478_{(0.035)}$	$0.478_{(0.034)}$	$0.476_{(0.036)}$
		SheepDog [11]	0.875	0.876	0.877	0.875	0.476	0.431	0.476	0.374	$0.493_{(0.007)}$	$0.438_{(0.077)}$	$0.493_{(0.007)}$	$0.360_{(0.021)}$
	(iv)	LLM Dir. [1]	0.675	0.779	0.690	0.652	0.532	0.675	0.532	0.412	$0.537_{(0.024)}$	$0.572_{(0.045)}$	$0.537_{(0.024)}$	$0.472_{(0.031)}$
		LLM CoT [1]	0.592	0.601	0.597	0.589	0.364	0.360	0.364	0.360	$0.549_{(0.029)}$	$0.591_{(0.056)}$	$0.549_{(0.029)}$	$0.491_{(0.034)}$
GossipCop	(i)	DeClarE [7]	0.815	0.748	0.717	0.730	0.482	0.459	0.482	0.398	$0.543_{(0.006)}$	$0.555_{(0.008)}$	$0.543_{(0.006)}$	$0.516_{(0.010)}$
		MAC [10]	0.839	0.787	0.799	0.793	0.500	0.250	0.500	0.333	$0.497_{(0.033)}$	$0.489_{(0.039)}$	$0.495_{(0.033)}$	$0.476_{(0.047)}$
		GET [12]	0.852	0.813	0.785	0.797	0.516	0.522	0.516	0.482	$0.535_{(0.052)}$	$0.539_{(0.057)}$	$0.536_{(0.055)}$	$0.529_{(0.054)}$
		MUSER [5]	0.882	0.835	0.848	0.841	0.562	0.634	0.562	0.494	$0.554_{(0.036)}$	$0.557_{(0.039)}$	$0.554_{(0.036)}$	$0.548_{(0.036)}$
		CompareNet [3]	0.930	0.902	0.908	0.905	0.495	0.367	0.495	0.385	$0.515_{(0.011)}$	$0.442_{(0.136)}$	$0.515_{(0.011)}$	$0.421_{(0.062)}$
	(ii)	dEFEND [9]	0.942	0.923	0.916	0.919	0.478	0.416	0.478	0.360	$0.501_{(0.025)}$	$0.418_{(0.125)}$	$0.501_{(0.025)}$	$0.411_{(0.062)}$
		WSDMS [13]	0.951	0.938	0.927	0.932	0.482	0.472	0.482	0.430	$0.487_{(0.009)}$	$0.487_{(0.009)}$	$0.487_{(0.009)}$	$0.486_{(0.009)}$
	(iii)	GenFEND [6]	0.958	0.955	0.930	0.941	0.490	0.490	0.490	0.488	$0.490_{(0.013)}$	$0.489_{(0.014)}$	$0.490_{(0.013)}$	$0.487_{(0.016)}$
		SheepDog [11]	0.982	0.981	0.969	0.975	0.506	0.527	0.506	0.386	$0.519_{(0.036)}$	$0.514_{(0.051)}$	$0.519_{(0.036)}$	$0.452_{(0.073)}$
	(iv)	LLM Dir. [1]	0.780	0.818	0.780	0.702	0.532	0.675	0.532	0.412	$0.537_{(0.024)}$	$0.572_{(0.045)}$	$0.537_{(0.024)}$	$0.472_{(0.031)}$
		LLM CoT [1]	0.628	0.516	0.518	0.516	0.364	0.360	0.364	0.360	$0.549_{(0.029)}$	$0.591_{(0.056)}$	$0.549_{(0.029)}$	$0.491_{(0.034)}$

publication date, we use Qwen3-14B [14] to generate Google-style search queries from both article-level key claim and sentence-level claims. The prompts explicitly require the model to preserve names, locations, and dates, and to distinguish the publication date from event dates mentioned in the article.

Specifically, we first extract a single main claim and a list of distinct sub-claims from the article. The main claim is intended to summarize the article-level assertion, while the sub-claims decompose it into independently verifiable predicates. We then generate one Google-style keyword query from the main claim and additional queries from the sub-claims.

To increase coverage of sparse factual cues, we also generate sentence-level queries. We split each article into sentences, remove short or boilerplate sentences, and greedily select up to ten sentences that add the most new named entities, numbers, or quoted strings. Each selected sentence is first rewritten into a self-contained claim using the full article context, resolving pronouns and relative time expressions where possible. We then generate two to four queries per claim, covering different applicable search angles such as the main event, involved entities, official records or statistics, and related background.

The final query set for an article is obtained by merging article-level and sentence-level queries. We remove exact duplicates case-insensitively and apply semantic deduplication within query-type groups using cosine similarity, with a threshold of 0.85; when two queries are near-duplicates, we keep the longer and more specific one. Finally, long queries are rewritten into compact keyword queries using KeyBERT [2], extracting the top five unigram-to-trigram keyphrases and removing repeated words. These final queries are used for Google Search corpus construction. After obtaining all searched results, we split each retrieved document into 100-word chunks with a stride of 50 words, yielding approximately 130 documents and 2,200 chunks per instance on average, which serve as retrieval units for BM25 [8].

to $G \times B$ online search calls per instance per update if live search is used. Since common search APIs charge per query (e.g., Google Custom Search charges $5 per 1,000 additional queries), we use online search only for corpus construction and perform verifier retrieval over the fixed instance-specific corpus with BM25.

Table 2: Statistics of the BreakNews imbalanced test sets.

Setting	Real	Fake	Total
85% Real	1,155	203	1,358
90% Real	1,155	128	1,283
95% Real	1,155	60	1,215

C Additional Experiment Settings & Results

C.1 Robustness under Class Imbalance

C.1.1 Datasets. Table 2 reports the statistics of all imbalanced test sets.

C.1.2 Results. Tables 3, 4, and 5 provide full class-wise results for the imbalanced BreakNews test sets. They complement Figure 4 by showing how each method trades off fake recall, fake precision, and real recall under increasingly skewed priors.

C.2 Complete Bidirectional Cross-Corpus Results

Figure 1 reports all nine supervised baselines and VeriFocus for the four transfer directions summarized in Figure 3. Models are trained

Table 3: Performance under the 85%:15% imbalanced setting. Mean and standard deviation are reported.

Model	Method	Macro-F1	MCC	Fake Class			Real Class		
				Prec	Recall	F1	Prec	Recall	F1
Zero-shot									
Ministral-3-8B	Direct	$0.459_{(0.031)}$	$-0.064_{(0.064)}$	$0.102_{(0.027)}$	$0.090_{(0.054)}$	$0.084_{(0.033)}$	$0.838_{(0.016)}$	$0.837_{(0.134)}$	$0.834_{(0.076)}$
	CoT	$0.475_{(0.015)}$	$0.108_{(0.028)}$	$0.188_{(0.011)}$	$0.626_{(0.025)}$	$0.289_{(0.016)}$	$0.889_{(0.009)}$	$0.525_{(0.016)}$	$0.660_{(0.015)}$
Llama-3.1-8B	Direct	$0.497_{(0.024)}$	$0.035_{(0.033)}$	$0.204_{(0.039)}$	$0.059_{(0.043)}$	$0.089_{(0.054)}$	$0.853_{(0.003)}$	$0.963_{(0.018)}$	$0.905_{(0.006)}$
	CoT	$0.528_{(0.019)}$	$0.086_{(0.029)}$	$0.199_{(0.019)}$	$0.374_{(0.013)}$	$0.260_{(0.017)}$	$0.870_{(0.005)}$	$0.734_{(0.032)}$	$0.796_{(0.021)}$
Qwen3-8B	Direct	$0.607_{(0.027)}$	$0.252_{(0.023)}$	$0.481_{(0.049)}$	$0.228_{(0.074)}$	$0.301_{(0.060)}$	$0.876_{(0.008)}$	$0.955_{(0.022)}$	$0.913_{(0.006)}$
	CoT	$0.626_{(0.016)}$	$0.275_{(0.025)}$	$0.324_{(0.025)}$	$0.521_{(0.014)}$	$0.399_{(0.019)}$	$0.906_{(0.003)}$	$0.808_{(0.023)}$	$0.854_{(0.013)}$
GPT-5.2	Direct	$0.659_{(0.002)}$	$0.324_{(0.003)}$	$0.386_{(0.003)}$	$0.496_{(0.006)}$	$0.434_{(0.003)}$	$0.907_{(0.001)}$	$0.862_{(0.002)}$	$0.884_{(0.001)}$
	CoT	$0.628_{(0.011)}$	$0.296_{(0.024)}$	$0.318_{(0.012)}$	$0.599_{(0.030)}$	$0.416_{(0.018)}$	$0.917_{(0.006)}$	$0.775_{(0.002)}$	$0.840_{(0.004)}$
GPT-5.2-RAG	ICL-Direct	$0.619_{(0.006)}$	$0.287_{(0.010)}$	$0.307_{(0.007)}$	$0.611_{(0.009)}$	$0.408_{(0.008)}$	$0.917_{(0.002)}$	$0.757_{(0.005)}$	$0.830_{(0.004)}$
	ICL-CoT	$0.534_{(0.004)}$	$0.214_{(0.007)}$	$0.233_{(0.003)}$	$0.711_{(0.008)}$	$0.351_{(0.004)}$	$0.921_{(0.002)}$	$0.588_{(0.005)}$	$0.718_{(0.004)}$
Few-shot / trained									
GPT-5.2	ICL-Direct	$0.646_{(0.025)}$	$0.310_{(0.042)}$	$0.361_{(0.051)}$	$0.525_{(0.072)}$	$0.424_{(0.033)}$	$0.909_{(0.009)}$	$0.832_{(0.049)}$	$0.868_{(0.025)}$
	ICL-CoT	$0.634_{(0.013)}$	$0.297_{(0.015)}$	$0.331_{(0.024)}$	$0.563_{(0.042)}$	$0.416_{(0.011)}$	$0.912_{(0.005)}$	$0.798_{(0.034)}$	$0.851_{(0.017)}$
GPT-5.2-RAG	ICL-Direct	$0.634_{(0.008)}$	$0.290_{(0.012)}$	$0.334_{(0.015)}$	$0.534_{(0.037)}$	$0.410_{(0.009)}$	$0.908_{(0.005)}$	$0.812_{(0.023)}$	$0.858_{(0.011)}$
	ICL-CoT	$0.601_{(0.022)}$	$0.286_{(0.014)}$	$0.289_{(0.020)}$	$0.677_{(0.052)}$	$0.404_{(0.013)}$	$0.926_{(0.006)}$	$0.704_{(0.053)}$	$0.799_{(0.032)}$
Ministral-3-8B	SFT	$0.589_{(0.025)}$	$0.197_{(0.033)}$	$0.281_{(0.038)}$	$0.429_{(0.049)}$	$0.337_{(0.020)}$	$0.891_{(0.002)}$	$0.802_{(0.059)}$	$0.843_{(0.033)}$
Llama-3.1-8B	SFT	$0.617_{(0.002)}$	$0.238_{(0.003)}$	$0.330_{(0.008)}$	$0.399_{(0.021)}$	$0.361_{(0.005)}$	$0.890_{(0.000)}$	$0.857_{(0.013)}$	$0.873_{(0.007)}$
Qwen3-8B	SFT	$0.446_{(0.019)}$	$0.205_{(0.008)}$	$0.204_{(0.006)}$	$0.875_{(0.017)}$	$0.331_{(0.006)}$	$0.948_{(0.003)}$	$0.400_{(0.032)}$	$0.562_{(0.031)}$
Qwen3-8B	Search-R1	$0.614_{(0.038)}$	$0.243_{(0.067)}$	$0.352_{(0.057)}$	$0.500_{(0.017)}$	$0.413_{(0.045)}$	$0.864_{(0.011)}$	$0.774_{(0.048)}$	$0.816_{(0.032)}$
Qwen3-8B	SearchAgent	$0.655_{(0.009)}$	$0.338_{(0.011)}$	$0.397_{(0.020)}$	$0.624_{(0.057)}$	$0.484_{(0.007)}$	$0.894_{(0.010)}$	$0.769_{(0.038)}$	$0.826_{(0.018)}$
Qwen3-8B	VeriFocus	$0.663_{(0.007)}$	$0.357_{(0.022)}$	$0.404_{(0.006)}$	$0.650_{(0.040)}$	$0.498_{(0.016)}$	$0.900_{(0.009)}$	$0.768_{(0.011)}$	$0.829_{(0.003)}$

Table 4: Performance under the 90%:10% imbalanced setting. Mean and standard deviation are reported.

Model	Method	Macro-F1	MCC	Fake Class			Real Class		
				Prec	Recall	F1	Prec	Recall	F1
Zero-shot									
Ministral-3-8B	Direct	$0.463_{(0.114)}$	$-0.054_{(0.013)}$	$0.064_{(0.015)}$	$0.089_{(0.058)}$	$0.065_{(0.011)}$	$0.892_{(0.028)}$	$0.840_{(0.133)}$	$0.862_{(0.076)}$
	CoT	$0.440_{(0.027)}$	$0.096_{(0.056)}$	$0.129_{(0.018)}$	$0.641_{(0.070)}$	$0.215_{(0.029)}$	$0.928_{(0.016)}$	$0.519_{(0.025)}$	$0.666_{(0.024)}$
Llama-3.1-8B	Direct	$0.507_{(0.015)}$	$0.032_{(0.019)}$	$0.146_{(0.017)}$	$0.060_{(0.036)}$	$0.082_{(0.037)}$	$0.902_{(0.002)}$	$0.963_{(0.018)}$	$0.931_{(0.008)}$
	CoT	$0.540_{(0.058)}$	$0.048_{(0.042)}$	$0.124_{(0.021)}$	$0.339_{(0.037)}$	$0.181_{(0.028)}$	$0.909_{(0.007)}$	$0.732_{(0.025)}$	$0.811_{(0.018)}$
Qwen3-8B	Direct	$0.593_{(0.019)}$	$0.204_{(0.018)}$	$0.348_{(0.045)}$	$0.208_{(0.070)}$	$0.252_{(0.045)}$	$0.916_{(0.005)}$	$0.955_{(0.022)}$	$0.935_{(0.008)}$
	CoT	$0.599_{(0.018)}$	$0.242_{(0.032)}$	$0.237_{(0.023)}$	$0.518_{(0.030)}$	$0.325_{(0.025)}$	$0.938_{(0.004)}$	$0.815_{(0.017)}$	$0.872_{(0.011)}$
GPT-5.2	Direct	$0.611_{(0.005)}$	$0.243_{(0.011)}$	$0.259_{(0.007)}$	$0.448_{(0.020)}$	$0.328_{(0.009)}$	$0.933_{(0.002)}$	$0.858_{(0.007)}$	$0.894_{(0.003)}$
	CoT	$0.578_{(0.004)}$	$0.225_{(0.007)}$	$0.213_{(0.004)}$	$0.557_{(0.016)}$	$0.309_{(0.005)}$	$0.940_{(0.002)}$	$0.772_{(0.008)}$	$0.848_{(0.005)}$
GPT-5.2-RAG	ICL-Direct	$0.569_{(0.007)}$	$0.215_{(0.017)}$	$0.204_{(0.008)}$	$0.563_{(0.023)}$	$0.300_{(0.012)}$	$0.940_{(0.003)}$	$0.757_{(0.005)}$	$0.839_{(0.004)}$
	ICL-CoT	$0.490_{(0.005)}$	$0.167_{(0.003)}$	$0.156_{(0.010)}$	$0.688_{(0.005)}$	$0.255_{(0.004)}$	$0.944_{(0.005)}$	$0.588_{(0.005)}$	$0.725_{(0.014)}$
Few-shot / trained									
GPT-5.2	ICL-Direct	$0.609_{(0.023)}$	$0.256_{(0.036)}$	$0.256_{(0.037)}$	$0.508_{(0.089)}$	$0.337_{(0.029)}$	$0.939_{(0.008)}$	$0.832_{(0.049)}$	$0.882_{(0.026)}$
	ICL-CoT	$0.590_{(0.017)}$	$0.233_{(0.018)}$	$0.227_{(0.021)}$	$0.529_{(0.035)}$	$0.317_{(0.016)}$	$0.939_{(0.002)}$	$0.798_{(0.034)}$	$0.862_{(0.019)}$
GPT-5.2-RAG	ICL-Direct	$0.592_{(0.005)}$	$0.228_{(0.010)}$	$0.229_{(0.008)}$	$0.503_{(0.046)}$	$0.314_{(0.006)}$	$0.937_{(0.004)}$	$0.813_{(0.023)}$	$0.870_{(0.012)}$
	ICL-CoT	$0.555_{(0.021)}$	$0.227_{(0.006)}$	$0.197_{(0.012)}$	$0.651_{(0.063)}$	$0.302_{(0.009)}$	$0.948_{(0.005)}$	$0.704_{(0.053)}$	$0.807_{(0.033)}$
Ministral-3-8B	SFT	$0.578_{(0.029)}$	$0.203_{(0.032)}$	$0.216_{(0.032)}$	$0.479_{(0.050)}$	$0.295_{(0.026)}$	$0.933_{(0.003)}$	$0.801_{(0.058)}$	$0.861_{(0.034)}$
Llama-3.1-8B	SFT	$0.607_{(0.002)}$	$0.233_{(0.008)}$	$0.254_{(0.004)}$	$0.435_{(0.030)}$	$0.320_{(0.006)}$	$0.932_{(0.002)}$	$0.858_{(0.012)}$	$0.893_{(0.005)}$
Qwen3-8B	SFT	$0.405_{(0.019)}$	$0.182_{(0.007)}$	$0.142_{(0.004)}$	$0.893_{(0.016)}$	$0.245_{(0.006)}$	$0.971_{(0.002)}$	$0.400_{(0.032)}$	$0.566_{(0.031)}$
Qwen3-8B	Search-R1	$0.578_{(0.014)}$	$0.185_{(0.011)}$	$0.246_{(0.020)}$	$0.440_{(0.041)}$	$0.314_{(0.006)}$	$0.902_{(0.003)}$	$0.791_{(0.040)}$	$0.842_{(0.022)}$
Qwen3-8B	SearchAgent	$0.613_{(0.016)}$	$0.278_{(0.023)}$	$0.287_{(0.019)}$	$0.592_{(0.060)}$	$0.385_{(0.017)}$	$0.925_{(0.007)}$	$0.772_{(0.040)}$	$0.841_{(0.022)}$
Qwen3-8B	VeriFocus	$0.616_{(0.035)}$	$0.288_{(0.074)}$	$0.287_{(0.039)}$	$0.620_{(0.086)}$	$0.392_{(0.054)}$	$0.929_{(0.016)}$	$0.764_{(0.014)}$	$0.839_{(0.014)}$

Table 5: Performance under the 95%:5% imbalanced setting. Mean and standard deviation are reported.

Model	Method	Macro-F1	MCC	Fake Class			Real Class		
				Prec	Recall	F1	Prec	Recall	F1
Zero-shot									
Ministral-3-8B	Direct	$0.555_{(0.008)}$	$0.109_{(0.016)}$	$0.155_{(0.015)}$	$0.150_{(0.017)}$	$0.152_{(0.015)}$	$0.956_{(0.001)}$	$0.958_{(0.003)}$	$0.957_{(0.002)}$
	CoT	$0.394_{(0.020)}$	$0.063_{(0.039)}$	$0.064_{(0.009)}$	$0.628_{(0.063)}$	$0.115_{(0.017)}$	$0.964_{(0.007)}$	$0.517_{(0.026)}$	$0.673_{(0.024)}$
Llama-3.1-8B	Direct	$0.519_{(0.012)}$	$0.042_{(0.023)}$	$0.093_{(0.014)}$	$0.078_{(0.048)}$	$0.082_{(0.032)}$	$0.953_{(0.002)}$	$0.962_{(0.018)}$	$0.957_{(0.008)}$
	CoT	$0.461_{(0.003)}$	$0.017_{(0.022)}$	$0.055_{(0.008)}$	$0.300_{(0.060)}$	$0.093_{(0.014)}$	$0.953_{(0.003)}$	$0.735_{(0.015)}$	$0.830_{(0.009)}$
Qwen3-8B	Direct	$0.568_{(0.009)}$	$0.141_{(0.022)}$	$0.186_{(0.037)}$	$0.189_{(0.069)}$	$0.180_{(0.025)}$	$0.958_{(0.003)}$	$0.955_{(0.022)}$	$0.956_{(0.010)}$
	CoT	$0.524_{(0.014)}$	$0.123_{(0.023)}$	$0.103_{(0.013)}$	$0.411_{(0.019)}$	$0.165_{(0.018)}$	$0.964_{(0.002)}$	$0.814_{(0.016)}$	$0.882_{(0.010)}$
GPT-5.2	Direct	$0.558_{(0.009)}$	$0.165_{(0.023)}$	$0.135_{(0.011)}$	$0.406_{(0.039)}$	$0.203_{(0.018)}$	$0.966_{(0.002)}$	$0.866_{(0.001)}$	$0.913_{(0.001)}$
	CoT	$0.531_{(0.002)}$	$0.177_{(0.008)}$	$0.118_{(0.003)}$	$0.567_{(0.017)}$	$0.196_{(0.005)}$	$0.972_{(0.001)}$	$0.781_{(0.001)}$	$0.866_{(0.000)}$
GPT-5.2-RAG	Direct	$0.515_{(0.007)}$	$0.155_{(0.019)}$	$0.106_{(0.007)}$	$0.556_{(0.035)}$	$0.178_{(0.012)}$	$0.970_{(0.002)}$	$0.757_{(0.005)}$	$0.851_{(0.004)}$
	CoT	$0.436_{(0.003)}$	$0.112_{(0.008)}$	$0.078_{(0.002)}$	$0.667_{(0.017)}$	$0.139_{(0.004)}$	$0.971_{(0.001)}$	$0.588_{(0.005)}$	$0.733_{(0.004)}$
Few-shot / trained									
GPT-5.2	ICL-Direct	$0.546_{(0.018)}$	$0.166_{(0.026)}$	$0.127_{(0.017)}$	$0.461_{(0.106)}$	$0.197_{(0.018)}$	$0.968_{(0.005)}$	$0.832_{(0.049)}$	$0.894_{(0.027)}$
	ICL-CoT	$0.530_{(0.010)}$	$0.157_{(0.017)}$	$0.114_{(0.005)}$	$0.500_{(0.076)}$	$0.186_{(0.008)}$	$0.969_{(0.003)}$	$0.798_{(0.034)}$	$0.875_{(0.019)}$
GPT-5.2-RAG	ICL-Direct	$0.542_{(0.008)}$	$0.174_{(0.005)}$	$0.124_{(0.005)}$	$0.511_{(0.042)}$	$0.200_{(0.004)}$	$0.970_{(0.002)}$	$0.813_{(0.023)}$	$0.884_{(0.013)}$
	ICL-CoT	$0.505_{(0.024)}$	$0.192_{(0.008)}$	$0.111_{(0.010)}$	$0.706_{(0.054)}$	$0.192_{(0.013)}$	$0.979_{(0.002)}$	$0.704_{(0.053)}$	$0.818_{(0.035)}$
Ministral-3-8B	SFT	$0.527_{(0.028)}$	$0.142_{(0.025)}$	$0.111_{(0.018)}$	$0.461_{(0.054)}$	$0.178_{(0.022)}$	$0.966_{(0.002)}$	$0.802_{(0.059)}$	$0.876_{(0.036)}$
Llama-3.1-8B	SFT	$0.542_{(0.004)}$	$0.136_{(0.002)}$	$0.118_{(0.002)}$	$0.372_{(0.025)}$	$0.179_{(0.001)}$	$0.963_{(0.001)}$	$0.855_{(0.013)}$	$0.906_{(0.007)}$
Qwen3-8B	SFT	$0.348_{(0.018)}$	$0.118_{(0.009)}$	$0.070_{(0.003)}$	$0.867_{(0.017)}$	$0.129_{(0.005)}$	$0.983_{(0.002)}$	$0.399_{(0.032)}$	$0.567_{(0.032)}$
Qwen3-8B	Search-R1	$0.515_{(0.017)}$	$0.105_{(0.000)}$	$0.115_{(0.008)}$	$0.400_{(0.071)}$	$0.177_{(0.002)}$	$0.948_{(0.002)}$	$0.777_{(0.056)}$	$0.853_{(0.033)}$
Qwen3-8B	SearchAgent	$0.543_{(0.015)}$	$0.194_{(0.001)}$	$0.149_{(0.009)}$	$0.580_{(0.056)}$	$0.237_{(0.006)}$	$0.962_{(0.003)}$	$0.760_{(0.040)}$	$0.848_{(0.024)}$
Qwen3-8B	VeriFocus	$0.544_{(0.028)}$	$0.193_{(0.070)}$	$0.149_{(0.030)}$	$0.572_{(0.121)}$	$0.236_{(0.048)}$	$0.961_{(0.011)}$	$0.766_{(0.013)}$	$0.852_{(0.010)}$

on the source corpus and evaluated directly on the target corpus without target-domain fine-tuning. The full VeriFocus framework achieves the best Macro-F1 among the compared methods in each direction.

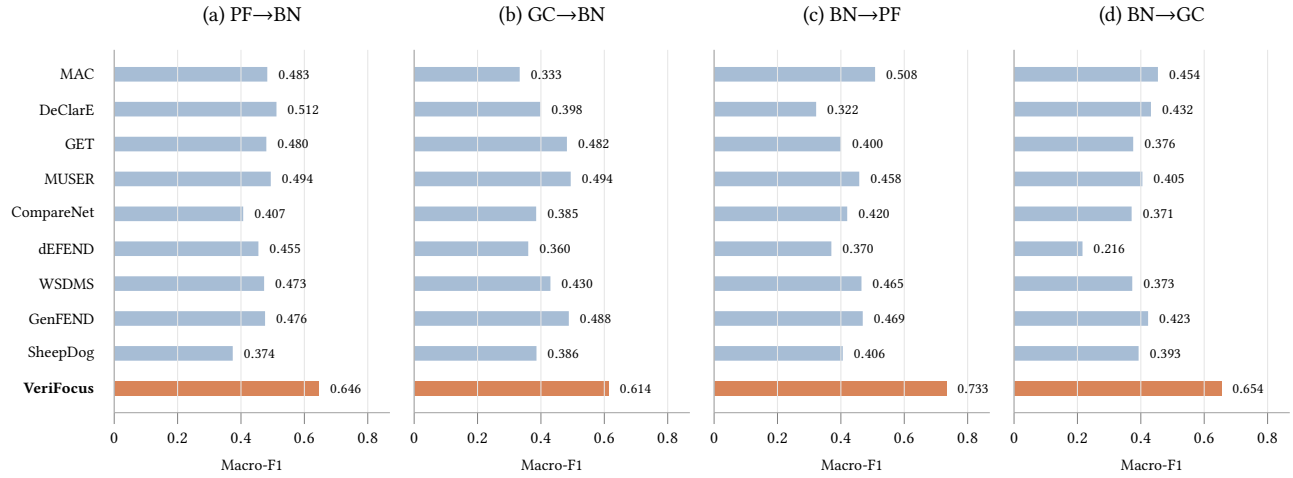


Figure 1: Complete cross-corpus Macro-F1 results. PF: PolitiFact; GC: GossipCop; BN: BreakNews. Each panel specifies the training corpus → test corpus. Models are evaluated without target-domain fine-tuning. VeriFocus is highlighted in orange.

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